

Interacting Multipoles of Rank 3 and 1:

$$-\frac{1}{15} \cdot T_{\alpha\beta\gamma\delta} \Omega_{\alpha\beta\gamma}^A \mu_\delta^B = R_z^{-9} \cdot -\frac{1}{15} \cdot \left[\begin{array}{l} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\delta^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \delta(\gamma, \delta) \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\delta^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \delta(\beta, \delta) \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\delta^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \delta(\beta, \gamma) \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\delta^B \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \delta(\alpha, \delta) \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\delta^B \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \delta(\alpha, \gamma) \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\delta^B \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \beta) \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\delta^B \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \beta) \cdot \delta(\gamma, \delta) \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\delta^B \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \gamma) \cdot \delta(\beta, \delta) \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\delta^B \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \delta) \cdot \delta(\beta, \gamma) \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\delta^B \end{array} \right] \quad (1)$$

Simplify deltas:

$$-\frac{1}{15} \cdot T_{\alpha\beta\gamma\delta} \Omega_{\alpha\beta\gamma}^A \mu_\delta^B = R_z^{-9} \cdot -\frac{1}{15} \cdot \left[\begin{array}{l} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\delta^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \delta(\gamma, \gamma) \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\gamma^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \delta(\beta, \beta) \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\beta^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \delta(\beta, \beta) \cdot \Omega_{\alpha\beta\beta}^A \cdot \mu_\delta^B \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \delta(\alpha, \alpha) \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\alpha^B \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \Omega_{\alpha\alpha\beta}^A \cdot \mu_\delta^B \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \delta(\alpha, \alpha) \cdot \Omega_{\alpha\alpha\gamma}^A \cdot \mu_\delta^B \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Omega_{\alpha\alpha\beta}^A \cdot \mu_\beta^B \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Omega_{\alpha\alpha\beta}^A \cdot \mu_\beta^B \\ +3 \cdot R_z^4 \cdot \delta(\alpha, \alpha) \cdot \delta(\beta, \beta) \cdot \Omega_{\alpha\beta\beta}^A \cdot \mu_\alpha^B \end{array} \right] \quad (2a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-9} \cdot -\frac{1}{15} \cdot \left[\begin{array}{l} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\delta^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\gamma^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\beta^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\delta \cdot \Omega_{\alpha\beta\beta}^A \cdot \mu_\delta^B \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\alpha^B \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\delta \cdot \Omega_{\alpha\alpha\beta}^A \cdot \mu_\delta^B \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\alpha\gamma}^A \cdot \mu_\delta^B \\ +3 \cdot R_z^4 \cdot \Omega_{\alpha\alpha\beta}^A \cdot \mu_\beta^B \\ +3 \cdot R_z^4 \cdot \Omega_{\alpha\alpha\beta}^A \cdot \mu_\beta^B \\ +3 \cdot R_z^4 \cdot \Omega_{\alpha\beta\beta}^A \cdot \mu_\alpha^B \end{array} \right] \quad (2b)$$

$$\xrightarrow{\text{reduced indices}} R_z^{-9} \cdot \frac{1}{15} \cdot \left[\begin{array}{c} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\delta^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\gamma^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\beta^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \Omega_{\alpha\beta\beta}^A \cdot \mu_\gamma^B \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\alpha^B \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \Omega_{\alpha\alpha\beta}^A \cdot \mu_\gamma^B \\ -15 \cdot R_z^2 \cdot R_\gamma \cdot R_\beta \cdot \Omega_{\alpha\alpha\gamma}^A \cdot \mu_\beta^B \\ +3 \cdot R_z^4 \cdot \Omega_{\alpha\alpha\beta}^A \cdot \mu_\beta^B \\ +3 \cdot R_z^4 \cdot \Omega_{\alpha\alpha\beta}^A \cdot \mu_\beta^B \\ +3 \cdot R_z^4 \cdot \Omega_{\alpha\beta\beta}^A \cdot \mu_\alpha^B \end{array} \right] \quad (2c)$$

$$\xrightarrow{\text{added up}} R_z^{-9} \cdot -\frac{1}{15} \cdot \left[\begin{array}{l} +105 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\delta^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\gamma^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\beta^B \\ -15 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot \Omega_{\alpha\beta\beta}^A \cdot \mu_\gamma^B \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \Omega_{\alpha\beta\gamma}^A \cdot \mu_\alpha^B \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \Omega_{\alpha\alpha\beta}^A \cdot \mu_\gamma^B \\ -15 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot \Omega_{\alpha\alpha\gamma}^A \cdot \mu_\beta^B \\ +6 \cdot R_z^4 \cdot \Omega_{\alpha\alpha\beta}^A \cdot \mu_\beta^B \\ +3 \cdot R_z^4 \cdot \Omega_{\alpha\beta\beta}^A \cdot \mu_\alpha^B \end{array} \right] \quad (2d)$$

Simplify R:

$$-\frac{1}{15} \cdot T_{\alpha\beta\gamma\delta} \Omega_{\alpha\beta\gamma}^A \mu_\delta^B = R_z^{-9} \cdot -\frac{1}{15} \cdot \left[\begin{array}{l} +105 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zzzz}^A \cdot \mu_z^B \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Omega_{zzz\gamma}^A \cdot \mu_\gamma^B \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Omega_{zzz\beta}^A \cdot \mu_\beta^B \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Omega_{zz\beta\beta}^A \cdot \mu_z^B \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Omega_{zzz\alpha}^A \cdot \mu_\alpha^B \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\alpha}^A \cdot \mu_z^B \\ -15 \cdot R_z^2 \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\alpha}^A \cdot \mu_z^B \\ +6 \cdot R_z^4 \cdot \Omega_{\alpha\alpha\beta}^A \cdot \mu_\beta^B \\ +3 \cdot R_z^4 \cdot \Omega_{\alpha\beta\beta}^A \cdot \mu_\alpha^B \end{array} \right] \quad (3a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-9} \cdot -\frac{1}{15} \cdot \left[\begin{array}{l} -15 \cdot R_z^4 \cdot \Omega_{zzz\gamma}^A \cdot \mu_\gamma^B \\ -15 \cdot R_z^4 \cdot \Omega_{zzz\beta}^A \cdot \mu_\beta^B \\ -15 \cdot R_z^4 \cdot \Omega_{zzz\alpha}^A \cdot \mu_\alpha^B \\ +6 \cdot R_z^4 \cdot \Omega_{\alpha\alpha\beta}^A \cdot \mu_\beta^B \\ +3 \cdot R_z^4 \cdot \Omega_{\alpha\beta\beta}^A \cdot \mu_\alpha^B \end{array} \right] \quad (3b)$$

$$\xrightarrow{\text{reduced indices}} R_z^{-9} \cdot -\frac{1}{15} \cdot \left[\begin{array}{l} -15 \cdot R_z^4 \cdot \Omega_{zz\alpha}^A \cdot \mu_\alpha^B \\ -15 \cdot R_z^4 \cdot \Omega_{zz\alpha}^A \cdot \mu_\alpha^B \\ -15 \cdot R_z^4 \cdot \Omega_{zz\alpha}^A \cdot \mu_\alpha^B \\ +6 \cdot R_z^4 \cdot \Omega_{\alpha\alpha\beta}^A \cdot \mu_\beta^B \\ +3 \cdot R_z^4 \cdot \Omega_{\alpha\beta\beta}^A \cdot \mu_\alpha^B \end{array} \right] \quad (3c)$$

$$\xrightarrow{\text{added up}} R_z^{-9} \cdot -\frac{1}{15} \cdot \left[\begin{array}{l} -45 \cdot R_z^4 \cdot \Omega_{zz\alpha}^A \cdot \mu_\alpha^B \\ +6 \cdot R_z^4 \cdot \Omega_{\alpha\alpha\beta}^A \cdot \mu_\beta^B \\ +3 \cdot R_z^4 \cdot \Omega_{\alpha\beta\beta}^A \cdot \mu_\alpha^B \end{array} \right] \quad (3d)$$

Simplify Dipoles:

$$-\frac{1}{15} \cdot T_{\alpha\beta\gamma\delta} \Omega_{\alpha\beta\gamma}^A \mu_\delta^B = R_z^{-9} \cdot -\frac{1}{15} \cdot \left[\begin{array}{l} -45 \cdot R_z^4 \cdot \Omega_{yzz}^A \cdot \mu_y^B \\ +6 \cdot R_z^4 \cdot \Omega_{y\alpha\alpha}^A \cdot \mu_y^B \\ +3 \cdot R_z^4 \cdot \Omega_{y\beta\beta}^A \cdot \mu_y^B \end{array} \right] \quad (4a)$$

$$\xrightarrow{\text{reduced indices}} R_z^{-9} \cdot -\frac{1}{15} \cdot \left[\begin{array}{l} -45 \cdot R_z^4 \cdot \Omega_{yzz}^A \cdot \mu_y^B \\ +6 \cdot R_z^4 \cdot \Omega_{y\alpha\alpha}^A \cdot \mu_y^B \\ +3 \cdot R_z^4 \cdot \Omega_{y\alpha\alpha}^A \cdot \mu_y^B \end{array} \right] \quad (4b)$$

$$\xrightarrow{\text{added up}} R_z^{-9} \cdot -\frac{1}{15} \cdot \left[\begin{array}{l} -45 \cdot R_z^4 \cdot \Omega_{yzz}^A \cdot \mu_y^B \\ +9 \cdot R_z^4 \cdot \Omega_{y\alpha\alpha}^A \cdot \mu_y^B \end{array} \right] \quad (4c)$$

Exploit Traceless Conditions:

$$-\frac{1}{15} \cdot T_{\alpha\beta\gamma\delta} \Omega_{\alpha\beta\gamma}^A \mu_{\delta}^B = R_z^{-9} \cdot -\frac{1}{15} \cdot [-45 \cdot R_z^4 \cdot \Omega_{yzz}^A \cdot \mu_y^B] \quad (5)$$

Combining everything:

$$-\frac{1}{15} \cdot T_{\alpha\beta\gamma\delta} \Omega_{\alpha\beta\gamma}^A \mu_{\delta}^B = [+3 \cdot R_z^{-5} \cdot \Omega_{yzz}^A \cdot \mu_y^B] \quad (6)$$